

CY 501P: Organic Chemistry Laboratory

Course Code	CY 501P
Course Name	Organic Chemistry Laboratory
L-T-P-C	0-0-8-4
Prerequisites	Core course
Intended for	M.Sc in Chemistry
Distribution	Core course
Approval	9th Senate (May 2016)

Course Contents

- Techniques [3 Lectures]
 - Learning structure elucidation of organic molecules & determination of following functional groups using IR and NMR spectroscopy:- (a) Amines (b) Imines (c) Esters (d) Amides (e) Carboxylic acid (f) Alcohol (g) Ethers (h) Aldehydes/Ketones [3 Lectures]
 - Extraction of Organic compounds & their characterization using IR, NMR and Mass spectroscopy:- (a) Extraction of caffeine from tea leaves. (i) Extraction of piperene from black pepper powder. [6 Lectures]
 - Quantitative Estimations:- Estimation of carbonyl group:- (i) Standardization of glucose solution by Fehlings Method. (ii) Determination of percentage purity of given sample of glucose. [3 Lectures]
 - Organic Synthesis:- Synthesis of organic molecules & their characterization using IR, NMR and Mass spectroscopy. (a) One step synthesis:- Synthesis of Bezpinacol (b) Multistep Synthesis:- Synthesis of Benzoic Acid; Benzaldehyde Benzoin Benzil Benzoic acid [6 Lectures]
 - Synthesis of a fluorogenic molecule and evaluation of its quantum yield. [4 Lectures]
(a) Introduction to c-c couplings. (b) Significance of c-c couplings in pharmaceutical industry. (c) Performing Sonogashira coupling at laboratory scale. [8 Lectures]
 - Synthesis of organic reagent, use of reagent in organic reaction & determination of its optical activity. (a) Synthesis of N-Bromosuccinimide (organic reagent, involves highly toxic and fuming Br₂ for synthesis). (b) Use of NBS in synthesis of halohydrin (Electrophilic Addition Reaction). (c) Determination of optical activity of generated product using polarimeter. [7 Lectures]
 - Use of Computer - Chem Draw, Draw the structure of simple aliphatic, aromatic, heterocyclic compounds with different substituents. Get the correct IUPAC name and predict the ¹H NMR signals. [1 Lectures]
 - Practical demonstration of instruments (a) HPLC (b) NMR [1 Lectures]
- NOTE- Alternate practical may be offered subject to conditions and availability of chemicals.

Books Recommended

- Vogels textbook of practical organic chemistry**, John Wiley and Sons.
- Vogels elementary quantitative organic analysis**, John Wiley and Sons.
- Robert, Gilbert & Rodewald, **An Introduction to Modern Experimental Organic Chemistry**, Saunders Publication.
- Skoog, West and Holler, **Fundamentals of Analytical Chemistry**, Saunders College publishing.
- Field, Sternhell & Kalman, **Organic structures from Spectra**, John Wiley and Sons.